# Customer Data Platform

## Single Design Document

### Hybrid Architecture & Data Ingestion

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# 1 Purpose of this Document

This Single Design Document provides a comprehensive architectural reference for the Customer Data Platform (CDP) operated by Example Organization. The CDP is a hybrid on-premises/cloud system that aggregates, cleanses, resolves, and serves customer data from 14 disparate source systems into a unified 360-degree customer view. This document captures the infrastructure design, data flows, high availability mechanisms, identity resolution logic, and operational procedures that govern the platform.

The Customer Data Platform is classified as a Tier-1 Business Critical system because it provides the authoritative source of customer identity and relationship data used by virtually every customer-facing application within the bank. When the CDP is unavailable or returns stale data, downstream systems including online banking, call center tools, fraud detection, and marketing automation all operate in a degraded state. The platform currently manages 82 million unique customer profiles and ingests 2.3 terabytes of new and updated data daily from across the enterprise.

## 1.1 Scope

This document covers the end-to-end architecture of the Customer Data Platform including on-premises data storage (Oracle Exadata RAC), change data capture (Oracle GoldenGate), cloud streaming infrastructure (Amazon MSK, Kinesis Data Analytics), data lake (S3 with Lake Formation), analytics warehouse (Amazon Redshift), and the customer API layer (OpenSearch Service). It also documents the identity resolution engine that merges records from multiple source systems into unified golden records, the data quality framework that ensures accuracy, and the security and compliance controls that protect sensitive customer information.

The scope explicitly excludes the individual source systems that feed data into the CDP, which are documented in their own respective SDDs. It also excludes downstream consumer applications that read from the CDP APIs, though their integration patterns and SLA requirements are referenced where relevant to architectural decisions.

## 1.2 Intended Audience

This document is intended for data platform engineers, cloud architects, security and compliance teams, database administrators, and any stakeholder requiring a detailed understanding of how customer data flows through the organization. It serves as the authoritative reference during architecture reviews, capacity planning exercises, compliance audits, and incident response activities involving customer data.

# 2 High Level Architecture

## 2.1 Hybrid Design Rationale

The Customer Data Platform employs a hybrid architecture with Oracle Exadata on-premises at DC-East serving as the system of record for customer profiles, and AWS cloud services providing scalable analytics, streaming, and API capabilities. This hybrid approach was chosen after extensive evaluation of fully on-premises and fully cloud-native alternatives, driven by three primary factors.

First, regulatory constraints require that the authoritative customer master data remain within the bank's physical data center infrastructure. The Office of the Comptroller of the Currency (OCC) and the bank's own data governance framework mandate that certain categories of customer PII be stored on infrastructure under direct bank control, with encryption keys managed by bank-operated hardware security modules. The Oracle Exadata deployment at DC-East satisfies these requirements while providing the performance characteristics necessary for real-time identity resolution.

Second, the analytics and API workloads exhibit highly variable demand patterns that are poorly served by fixed on-premises capacity. Marketing campaign activations can generate 10x normal API query volume within minutes, and quarterly regulatory reporting requires massive parallel scan operations that would be prohibitively expensive to provision for on-premises. AWS cloud services provide the elasticity to handle these burst workloads cost-effectively.

Third, the data lake and machine learning workloads benefit from the breadth of AWS managed services (Glue, SageMaker, Athena, Lake Formation) that would require significant engineering effort to replicate on-premises. By placing these workloads in the cloud while keeping the core profile store on-premises, the platform achieves both regulatory compliance and engineering velocity.

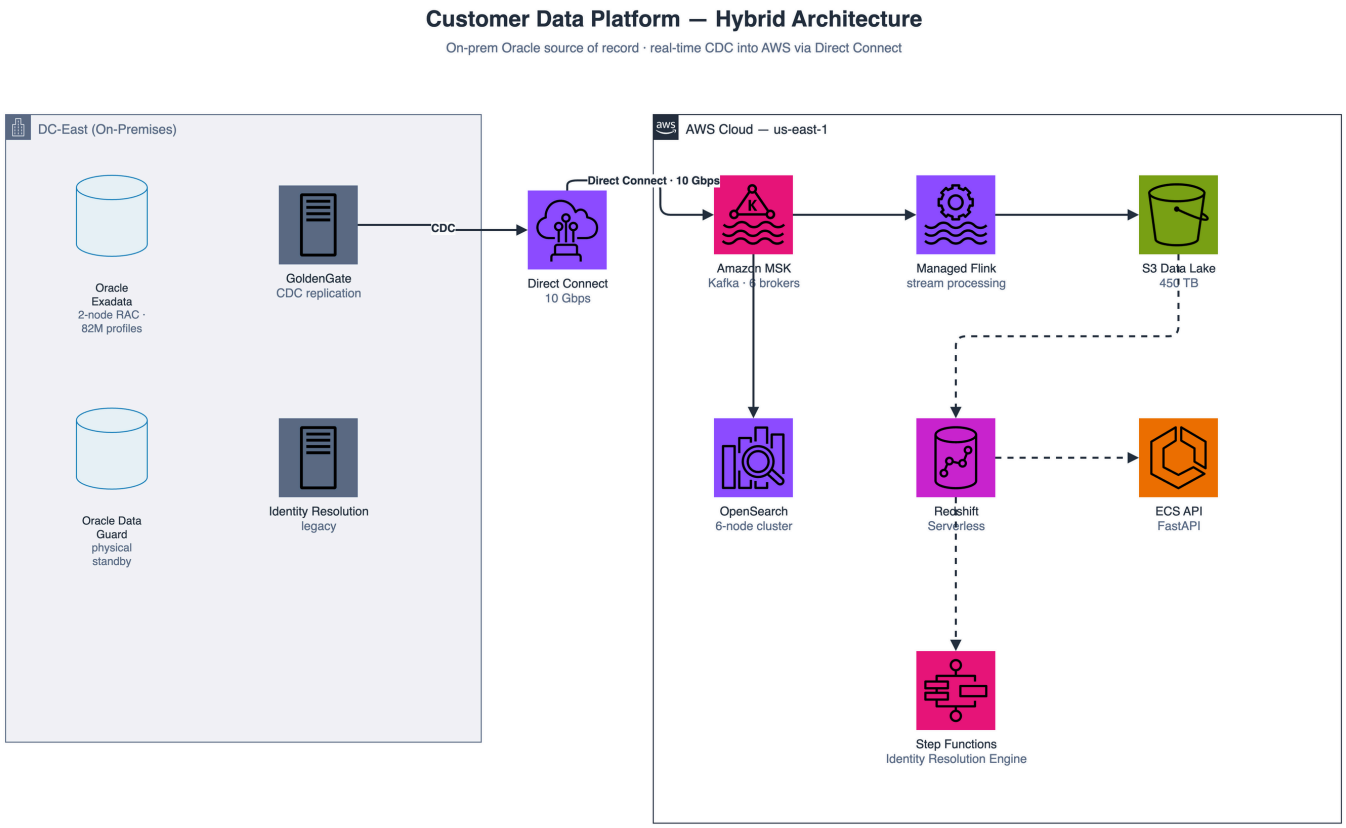



*Figure 1: Customer Data Platform - Hybrid Architecture Overview*

## 2.2 Architecture Principles

The CDP architecture is built upon the following foundational principles that guide all design decisions and trade-off resolutions:

**Single Source of Truth: Every customer has exactly one golden record in the CDP, identified by a globally unique Customer Master ID (CMID). All downstream systems reference this CMID rather than maintaining their own customer identity, eliminating the data silos and inconsistencies that previously existed across the bank's systems.**

**Real-Time by Default: Data changes in source systems are propagated to the CDP within seconds via change data capture, not hours via batch ETL. This ensures that customer-facing applications always have access to current information. Batch processing is reserved only for workloads that are inherently batch-oriented (regulatory reporting, model training).**

**Privacy by Design: Data classification, access control, and retention policies are enforced at the platform level rather than delegated to individual consuming applications. The CDP implements field-level encryption, dynamic data masking, and automated retention enforcement to ensure compliance with GDPR, CCPA, and internal data governance policies.**

**Resilience Over Performance: Where trade-offs exist between maximum throughput and system resilience, the architecture favors resilience. This manifests in choices such as synchronous replication for the profile store,**

**three-AZ deployment for all cloud components, and graceful degradation patterns that serve stale data rather than returning errors.**

## 2.3 System Context

The Customer Data Platform sits at the center of the organization's data ecosystem, receiving data from 14 upstream source systems and serving 47 downstream consumers. The source systems span the full range of banking operations: core banking (account and transaction data), card processing (Visa/Mastercard transaction streams), digital channels (online and mobile clickstream), CRM (Salesforce customer interactions), branch systems (teller transactions and document imaging), external data providers (credit bureau scores, address verification), and regulatory reporting systems.

Downstream consumers include the online banking platform, mobile application backend, call center customer view, marketing automation (Adobe Campaign), fraud detection engine, anti-money laundering system, credit decisioning platform, and the enterprise data warehouse. Each consumer accesses the CDP through one of three interfaces: the real-time Customer 360 API (OpenSearch-backed, sub-50ms P99), the batch export interface (S3-based Parquet files), or the analytics query interface (Redshift, for complex aggregations).

# 3 On-Premises Infrastructure

## 3.1 Oracle Exadata RAC

The on-premises foundation of the Customer Data Platform is an Oracle Exadata X9M Database Machine configured as a 4-node Real Application Clusters (RAC) deployment. Exadata was selected for its ability to deliver extreme I/O performance through Smart Scan offloading, hardware-accelerated compression, and persistent memory (PMEM) caching -- capabilities critical for the identity resolution engine's pattern matching workload that performs billions of comparisons per day.

The RAC cluster consists of four database nodes, each equipped with two 32-core Intel Xeon processors, 1.5 TB of RAM, and 12.8 TB of NVMe flash for local caching. The shared storage tier provides 200 TB of usable capacity across 14 storage cells with High Redundancy ASM diskgroups ensuring triple-mirror protection for the customer master data. Under normal operations, the cluster processes approximately 85,000 transactions per second with a P99 latency of 3.2 milliseconds for single-record lookups.

The RAC configuration provides both high availability and horizontal scalability. Any single node can fail without service interruption -- transparent application failover (TAF) redirects connections to surviving nodes within 8 seconds. For planned maintenance, rolling upgrades allow patching one node at a time while the remaining three nodes handle the full production workload. The cluster is currently operating at 58% average CPU utilization during peak hours, providing substantial headroom for growth.

Data Guard provides disaster recovery capability with a physical standby database maintained at DC-West. The standby receives redo logs in real-time via asynchronous shipping (ASYNC mode with a maximum lag of 15 seconds) and can be activated as the primary within 3 minutes in the event of a complete DC-East failure. The choice of ASYNC over SYNC mode was made to avoid imposing cross-data-center latency on every write operation, accepting the trade-off of potential data loss bounded by the replication lag.

## 3.2 Oracle GoldenGate CDC

Oracle GoldenGate 21c provides the change data capture (CDC) mechanism that propagates database changes from the Exadata cluster to downstream consumers in near-real-time. GoldenGate captures changes at the redo log level, providing a non-intrusive mechanism that adds negligible overhead to the source database (measured at less than 2% CPU impact on the Exadata nodes).

The GoldenGate deployment consists of two Extract processes (one primary, one for redundancy), a Trail File system on dedicated NVMe storage for buffering, and four Replicat processes that deliver changes to different targets. The primary Replicat delivers to Amazon MSK (Kafka) for real-time streaming consumers. A secondary Replicat maintains the Data Guard standby. A third feeds the identity resolution engine's incremental processing queue. The fourth provides audit-grade change logging to immutable S3 storage for compliance purposes.

GoldenGate is configured with Integrated Extract mode, which leverages the Oracle logminer infrastructure for maximum reliability and supportability. The Extract process captures all DML operations (INSERT, UPDATE, DELETE) on the 47 tables that comprise the customer data model, transforming them into a canonical JSON format before publishing to the downstream Kafka topics. Average end-to-end latency from database commit to Kafka message availability is 1.8 seconds under normal load.

### 3.3 On-Premises Networking

The on-premises network infrastructure for the CDP uses a dedicated VLAN (VLAN 350, 10.2.35.0/24) within the data center's secure data zone. This VLAN is isolated from general application traffic and is accessible only from authorized source systems and the AWS Direct Connect termination point. Network segmentation is enforced by Palo Alto PA-7080 next-generation firewalls in an active-passive HA configuration.

The Exadata cluster uses a dedicated InfiniBand fabric for inter-node communication and storage access, providing 100 Gbps bandwidth between compute and storage cells. Client access is through a 25 Gbps Ethernet bonded interface using Oracle SCAN (Single Client Access Name) for load balancing across RAC instances. The GoldenGate servers connect to the Exadata via a dedicated 10 Gbps Ethernet link to ensure that CDC traffic does not compete with client query traffic.

# 4 Cloud Infrastructure

The cloud tier of the Customer Data Platform is deployed in AWS us-east-1 (N. Virginia), chosen for its proximity to DC-East (less than 5ms network latency via Direct Connect), its breadth of available services, and its compliance certifications (SOC 2 Type II, PCI-DSS Level 1, ISO 27001). All cloud components are deployed across three Availability Zones for high availability.

## 4.1 Amazon MSK (Kafka)

Amazon Managed Streaming for Apache Kafka (MSK) serves as the central nervous system of the CDP's real-time data pipeline. The cluster consists of 9 broker nodes (3 per AZ) running Apache Kafka 3.5.1, configured with 120 topics that correspond to the different data domains and processing stages within the platform.

The cluster is provisioned with kafka.m5.4xlarge instances providing 16 vCPUs and 64 GB RAM per broker. Storage is configured at 2 TB per broker using gp3 EBS volumes with provisioned IOPS (16,000 IOPS per volume). This configuration supports the current peak throughput of 2.3 TB/day with a replication factor of 3 and a retention period of 7 days, providing ample buffer for downstream consumer recovery scenarios.

Topic partitioning is designed around the Customer Master ID (CMID) as the partition key, ensuring that all events for a given customer are processed in order within a single partition. This ordering guarantee is critical for the identity resolution engine, which must process merge and split events in the correct sequence to maintain data integrity. Topics use 64 partitions by default, with high-volume topics (account transactions, clickstream) scaled to 128 partitions.

## 4.2 Kinesis Data Analytics (Flink)

Amazon Kinesis Data Analytics with Apache Flink provides the stream processing layer that transforms, enriches, and routes events in real-time. Three Flink applications run continuously, each handling a distinct processing domain: event enrichment (adding reference data to raw events), real-time aggregation (computing rolling metrics like transaction velocity per customer), and anomaly detection (flagging unusual patterns for the fraud team).

The Flink applications use a parallelism of 16 (distributed across multiple task managers) and leverage RocksDB state backend for maintaining windowed aggregations. Checkpointing is configured at 60-second intervals with exactly-once semantics, ensuring that no events are lost or duplicated even during application restarts or scaling events. The applications process an average of 28,000 events per second with a P99 processing latency of 340 milliseconds.

## 4.3 Amazon S3 Data Lake

The S3 Data Lake serves as the long-term storage tier and the foundation for batch analytics, machine learning model training, and regulatory reporting. Data is organized into a multi-layer architecture: raw (landing zone for unprocessed source data), curated (cleansed and standardized data), and consumption (pre-aggregated datasets optimized for specific use cases).

AWS Lake Formation provides the governance layer for the data lake, managing fine-grained access control at the column and row level. Lake Formation integrates with the bank's existing Active Directory groups to enforce data classification-based access policies. For example, PII columns (SSN, date of birth, full address) are accessible only to authorized roles, while non-sensitive analytics columns are broadly available to the data science team.

Data in the lake is stored in Apache Parquet format with Snappy compression, providing an average compression ratio of 8:1 relative to raw JSON inputs. Partitioning follows a date/source_system/data_domain hierarchy that optimizes both time-range queries and source-system-specific reprocessing scenarios. The total data lake volume is currently 745 TB across all layers, growing at approximately 35% year-over-year.

### 4.4 Amazon Redshift

Amazon Redshift serves as the analytical query engine for complex aggregations, cross-customer analytics, and regulatory reporting extracts. The cluster consists of 8 ra3.4xlarge nodes providing 768 GB total RAM and 512 TB of managed storage. Redshift Spectrum extends query capability to the full S3 data lake without requiring data movement, enabling ad-hoc analysis across the complete historical dataset.

The Redshift cluster is loaded via two mechanisms: real-time micro-batch loading from the Flink stream processing layer (every 5 minutes via COPY from S3 staging), and daily batch loads from the curated data lake layer for historical backfill and full-refresh materializations. Workload Management (WLM) configuration allocates separate queues for interactive queries (priority), scheduled reports (standard), and data loading (background), ensuring that analytical users experience consistent sub-10-second query response times.

### 4.5 OpenSearch Service

Amazon OpenSearch Service provides the real-time Customer 360 API backend, enabling sub-50-millisecond lookups of customer profiles by any combination of identifiers (name, address, phone, email, account number, SSN token). The cluster consists of 6 data nodes (r6g.2xlarge) across 3 AZs with 3 dedicated master nodes for cluster coordination.

The OpenSearch index stores a denormalized view of each customer's golden record including demographic information, account summary, channel preferences, and relationship metadata. Documents are updated in near-real-time via the Flink stream processing layer, with an average index-to-searchable latency of 2.3 seconds. The index currently contains 82 million documents (one per golden record) with an average document size of 4.2 KB.

The Customer 360 API serves approximately 12,000 requests per second during peak hours, primarily from the online banking platform, mobile application, and call center tools. The API supports both exact-match queries (O(1) lookup by CMID or account number) and fuzzy-match queries (probabilistic search by partial name and address) used by call center agents for customer identification.

# 5 High Availability Design

## 5.1 On-Premises HA

The on-premises tier achieves high availability through Oracle RAC's shared-nothing compute architecture with shared-everything storage. The 4-node RAC cluster can tolerate the simultaneous failure of any two nodes while still maintaining sufficient capacity to serve the full production workload. Oracle Clusterware manages automatic node failure detection (via CSS voting disk heartbeat at 1-second intervals) and initiates transparent application failover within 8 seconds of detecting a node loss.

Storage-level redundancy is provided by ASM High Redundancy diskgroups, which maintain three copies of every data block across different failure groups (mapped to different storage cells). This triple-mirror configuration survives the simultaneous failure of any two storage cells without data loss. Combined with the compute redundancy of RAC, the on-premises tier provides an effective availability of 99.99% (measured 99.995% over the trailing 12 months).

Oracle GoldenGate achieves HA through redundant Extract processes running on different RAC nodes. If the primary Extract process or its hosting node fails, the standby Extract automatically resumes capture from the last checkpointed position in the redo log. The maximum data loss during an Extract failover is zero (no events are lost) though there may be a brief period of increased latency (typically 5-10 seconds) while the standby process initializes.

## 5.2 Cloud HA

All AWS components are deployed across three Availability Zones to survive the complete loss of any single AZ without service interruption. Amazon MSK distributes Kafka brokers evenly across AZs with a replication factor of 3, ensuring that partition leaders are always available on at least two AZs. Topic partitions are automatically rebalanced when a broker becomes unavailable.

Amazon Redshift uses the ra3 node type with managed storage, which automatically replicates data to S3 for durability and enables automatic node replacement within minutes if a node failure occurs. The cluster maintains an automatic snapshot every 8 hours with a retention of 35 days, enabling point-in-time recovery to any 5-minute interval.

OpenSearch Service is configured with zone awareness enabled, distributing primary and replica shards across all three AZs. The cluster can lose an entire AZ (2 of 6 data nodes) while still serving all queries from the remaining nodes, though with slightly increased latency until the cluster rebalances. Auto-scaling is configured to add additional data nodes if average CPU exceeds 70% for 5 consecutive minutes.

## 5.3 Cross-Environment Resilience

The hybrid architecture introduces a cross-environment dependency: the cloud tier depends on the on-premises CDC feed for data freshness. If the Direct Connect link or the GoldenGate process becomes unavailable, the cloud tier continues to serve queries using its last-known-good data (stale but not failed). This graceful degradation approach means that a Direct Connect outage results in increasingly stale data in the cloud tier but does not cause errors or outages for downstream consumers.

Monitoring thresholds are configured to alert on data staleness: a warning alert fires when the CDC lag exceeds 30 seconds, and a critical alert fires at 5 minutes. The platform can tolerate up to 4 hours of CDC pipeline downtime before downstream consumers begin to report material data freshness issues (based on SLA agreements with consuming

applications). During extended outages, a batch catch-up mechanism can replay missed changes from the Oracle redo logs once connectivity is restored.

# 6 Data Ingestion Pipelines

## 6.1 Real-Time CDC Pipeline

The real-time data ingestion pipeline is the primary mechanism by which source system changes flow into the Customer Data Platform. The pipeline operates continuously, capturing database-level changes from the 14 source systems and delivering them to the appropriate processing stages within an average end-to-end latency of 4.2 seconds (from source database commit to availability in the Customer 360 API).

The CDC pipeline begins at Oracle GoldenGate, which captures changes from the Exadata redo logs and transforms them into a canonical Avro-encoded event format. Each event includes the full after-image of the changed row, the operation type (insert/update/delete), the source system identifier, a monotonically increasing sequence number, and a schema version identifier. Events are published to Amazon MSK topics organized by source system and data domain (e.g., core-banking.accounts, cards.transactions).

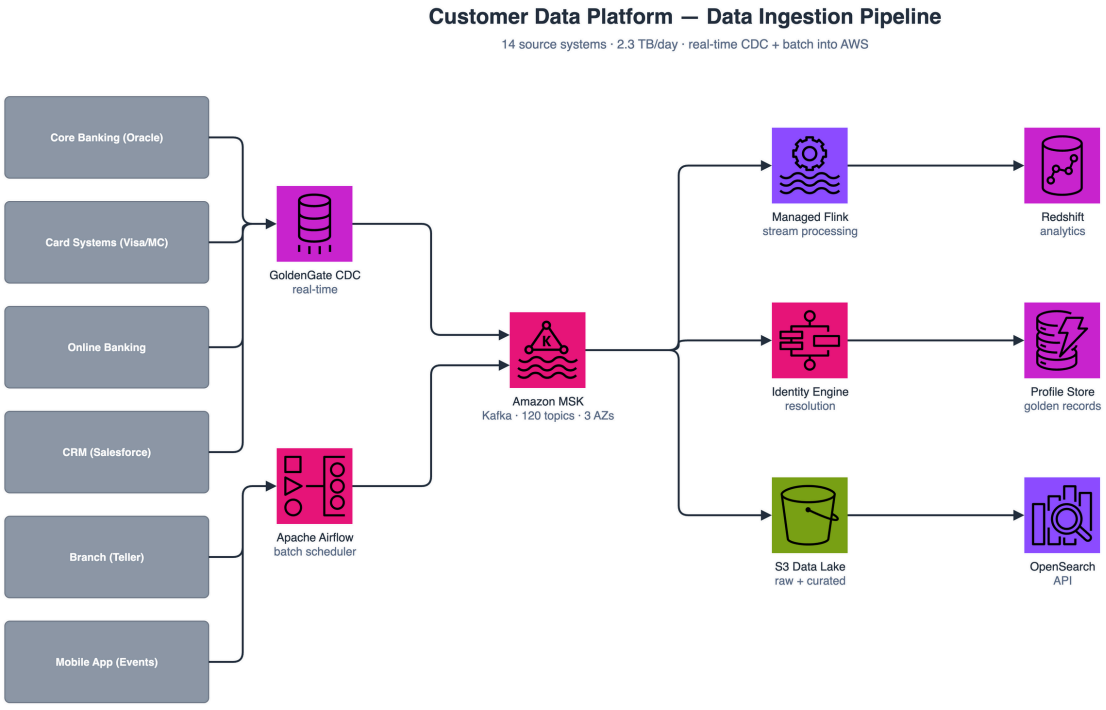



*Figure 2: Data Ingestion Pipeline Architecture*

From MSK, events flow through the Kinesis Flink stream processing layer where they undergo enrichment (joining with reference data such as product catalogs and branch information), validation (schema conformance and business rule checks), and routing (directing events to the appropriate downstream destinations based on their data domain and processing requirements).

The pipeline implements exactly-once delivery semantics through a combination of idempotent producers (GoldenGate to MSK), transactional consumers (Flink checkpoint coordination), and deduplication at the sink level (using the source sequence number as a natural idempotency key). This ensures that even in failure and recovery scenarios, no customer record is duplicated or lost in the downstream stores.

### 6.2 Batch Ingestion (Airflow)

While real-time CDC handles the majority of data ingestion, certain source systems and workloads are better served by batch processing. Apache Airflow (managed via Amazon MWAA) orchestrates 23 batch DAGs that run on defined schedules, handling workloads such as credit bureau file ingestion (daily), regulatory reporting extracts (daily and monthly), full-table reconciliation (weekly), and machine learning feature engineering (daily).

The batch pipeline follows an extract-load-transform (ELT) pattern: source data is extracted via JDBC or file transfer into the S3 raw layer, then transformed in-place using AWS Glue Spark jobs or Redshift stored procedures. This approach minimizes the load on source systems (extraction is a simple table scan or file copy) and leverages the scalable compute of the cloud tier for transformation logic.

Batch and real-time pipelines converge at the curated layer of the data lake, where both produce Parquet files conforming to the same schema. A reconciliation process runs daily to verify consistency between the real-time stream (which may have missed events during transient failures) and the batch full-extract (which provides a complete point-in-time snapshot). Discrepancies are automatically resolved by replaying the batch data for affected records.

### 6.3 Schema Evolution

Source system schemas evolve over time as business requirements change. The CDP uses Apache Avro with a schema registry (Confluent Schema Registry, self-hosted on ECS) to manage schema evolution. All schema changes must be backward-compatible: new fields are added with default values, field types are never narrowed, and fields are never removed (only deprecated). This ensures that downstream consumers can process both old and new format messages without coordination.

The schema registry enforces compatibility rules automatically at the producer level: GoldenGate's Avro formatter validates each schema change against compatibility rules before allowing publication. If a source system makes an incompatible schema change (such as renaming a column), the GoldenGate Extract process rejects the change and alerts the data engineering team, preventing downstream breakage. Over the past 12 months, the schema registry has blocked 7 incompatible changes that would have caused consumer failures.

# 7 Identity Resolution Engine

The Identity Resolution Engine is the intellectual core of the Customer Data Platform, responsible for determining which records from different source systems refer to the same real-world customer. The engine employs a four-stage matching pipeline that progressively applies more sophisticated (and computationally expensive) matching techniques to resolve customer identity with a measured accuracy of 99.97% and a false positive rate of 0.03%.

The engine processes both new records entering the platform (requiring assignment to an existing golden record or creation of a new one) and existing records that may need to be merged or split as new information becomes available. The current golden record population stands at 82 million unique customers, derived from approximately 221 million source system records (an average merge ratio of 2.7:1 indicating that each customer appears in an average of 2.7 source systems).

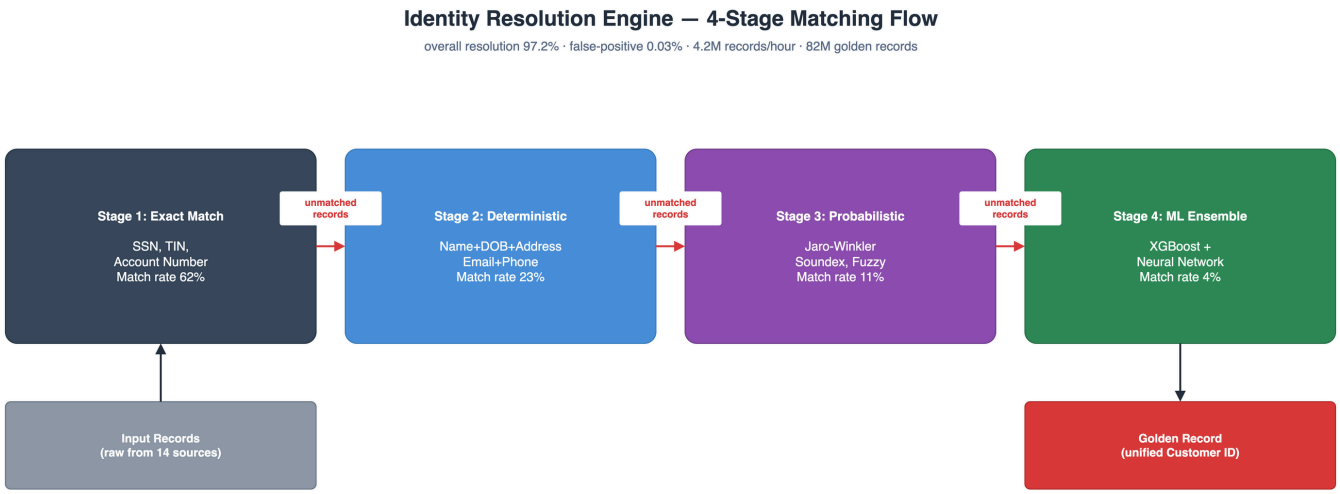



*Figure 3: Identity Resolution Engine - 4-Stage Matching Flow*

## 7.1 Deterministic Matching

The first two stages of the identity resolution pipeline use deterministic matching rules that produce binary (match/no-match) outcomes based on exact or near-exact attribute comparisons. Stage 1 (Exact Match) compares high-confidence unique identifiers: Social Security Number (SSN), Tax Identification Number (TIN), and account numbers that are known to be unique across systems. This stage resolves approximately 62% of all incoming records with zero false positives.

Stage 2 (Deterministic Composite) applies rules based on combinations of attributes that together provide high confidence: full name + date of birth + ZIP code, email address + last four of SSN, or phone number + full name. Each rule combination has been empirically validated against a labeled dataset of 500,000 manually-verified matches to confirm a false positive rate below 0.01%. This stage resolves an additional 23% of records.

## 7.2 Probabilistic Matching

Stage 3 (Probabilistic) handles the remaining 15% of records that could not be resolved deterministically, typically due to data quality issues such as misspelled names, outdated addresses, or transposed digits. This stage applies string similarity algorithms including Jaro-Winkler distance (for names), Soundex and Double Metaphone (for phonetic matching), and Levenshtein distance (for address components).

Each comparison produces a similarity score between 0 and 1, and scores across multiple attributes are combined using a weighted average derived from the Fellegi-Sunter probabilistic record linkage framework. Records are classified as matches (combined score above 0.92), non-matches (below 0.65), or uncertain (between 0.65 and 0.92). Uncertain records are escalated to Stage 4. This stage resolves approximately 11% of incoming records with a measured false positive rate of 0.08%, which is further reduced by the ML stage.

## 7.3 ML Ensemble Model

Stage 4 (ML Ensemble) is the final resolution stage, handling the 4% of records that remain unresolved after probabilistic matching. This stage uses a machine learning ensemble combining XGBoost (for structured feature comparisons) and a neural network model (for learning complex non-linear relationships between attributes). The ensemble was trained on 2.3 million manually-labeled record pairs and achieves an F1 score of 0.994 on the holdout test set.

The ML model considers over 180 features derived from the input records, including all attribute-level similarity scores from Stage 3, behavioral features (transaction pattern similarity, channel usage overlap), temporal features (account opening dates, address change dates), and network features (shared household indicators, authorized user relationships). The model produces a match probability that, combined with a tuned decision threshold of 0.87, achieves the overall 0.03% false positive rate target.

Records that the ML model cannot confidently classify (predicted probability between 0.70 and 0.87) are routed to a manual review queue staffed by the Data Quality team. Approximately 0.2% of all incoming records require manual review, representing roughly 4,600 records per day. Manual decisions are fed back into the model training pipeline as labeled examples, enabling continuous model improvement.

# 8 Network & Connectivity

## 8.1 AWS Direct Connect

Connectivity between DC-East and AWS us-east-1 is provided by two dedicated AWS Direct Connect circuits, each provisioned at 10 Gbps, from different telecommunications providers (Equinix and Megaport). The circuits terminate at different Direct Connect locations (EqDC2 and CoreSite NY1) to eliminate any single point of failure in the physical connectivity path.

Each circuit carries a Virtual Interface (VIF) configured as a private VIF with BGP peering to the bank's on-premises routers. BGP path prepending is used to designate the Equinix circuit as primary and the Megaport circuit as secondary, with automatic failover within 3 seconds of detecting a BGP session drop (BFD-accelerated). Under normal conditions, the primary circuit carries all production traffic while the secondary carries monitoring and management traffic.

The measured one-way latency via Direct Connect is 2.1 ms (Equinix path) and 2.4 ms (Megaport path), well within the 5ms budget required for the real-time CDC pipeline to meet its end-to-end latency SLA. Current peak bandwidth utilization is 4.8 Gbps (24% of aggregate capacity), providing substantial headroom for the projected 35% annual data growth. Encryption is applied at the application layer (TLS 1.3 for API traffic, Kafka TLS for streaming) rather than at the circuit level, avoiding the overhead of IPsec encryption on Direct Connect.

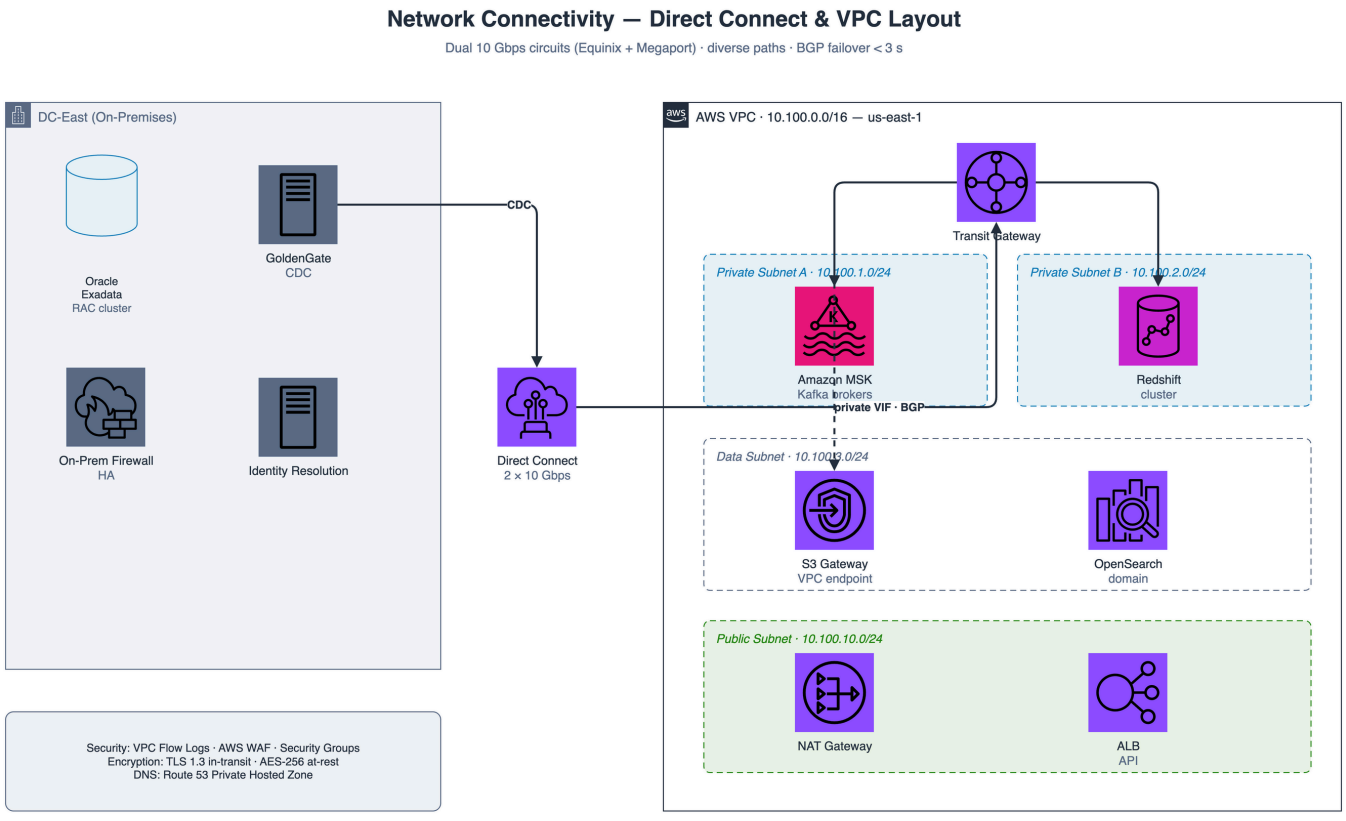



*Figure 4: Network Connectivity - Direct Connect & VPC Layout*

## 8.2 VPC Architecture

The AWS components reside in a dedicated VPC (10.100.0.0/16) with subnets organized by function and security tier. Private subnets (10.100.1.0/24 through 10.100.6.0/24) host all data processing components (MSK, Redshift, OpenSearch, Flink) with no direct internet access. A public subnet (10.100.0.0/24) hosts NAT Gateways and Application Load Balancers that serve the external-facing Customer 360 API.

Security Groups enforce least-privilege network access at the instance level. Each service component has a dedicated Security Group that allows inbound traffic only from known sources on specific ports. For example, the MSK Security Group allows inbound on port 9094 (TLS) only from the GoldenGate source IP range (on-premises) and the Flink task manager Security Group. No Security Group allows inbound traffic from 0.0.0.0/0.

VPC Endpoints (Gateway type for S3, Interface type for other services) ensure that traffic between the VPC and AWS services remains on the AWS backbone network rather than traversing the public internet. This provides both security benefits (traffic never leaves the AWS network) and performance benefits (lower latency and higher bandwidth than internet-routed paths).

# 9 Data Quality Framework

The Data Quality Framework ensures that data flowing through the Customer Data Platform meets defined accuracy, completeness, consistency, and timeliness standards. The framework operates at multiple points in the data lifecycle: at ingestion (validating incoming data against source system contracts), during processing (detecting anomalies and drift), and at consumption (monitoring SLA compliance for downstream consumers).

The framework is implemented using Great Expectations as the rule engine, integrated with the Airflow orchestration layer for batch workloads and with custom Flink operators for streaming workloads. Quality rules are defined as code (version-controlled in Git) and organized into expectation suites by data domain. Currently, 847 individual quality rules are in production across all data domains.

Quality metrics are continuously published to a dedicated monitoring dashboard that provides visibility into data health across all pipeline stages. Key metrics include record completeness (percentage of non-null values for required fields, current: 99.4%), referential integrity (percentage of foreign key relationships that resolve, current: 99.8%), freshness (time since last update, per source system), and uniqueness (absence of unintended duplicate records, monitored via daily reconciliation).

When a quality rule violation is detected, the framework follows a tiered response model. Minor violations (individual record issues) are logged and the record is routed to a quarantine queue for manual review while processing continues for all other records. Major violations (systemic issues affecting more than 1% of records in a batch) trigger a pipeline halt and an immediate alert to the data engineering team. Critical violations (data loss or corruption detected) trigger a pipeline halt, alert, and automatic rollback to the last known-good state.

# 10 Security, Privacy & Compliance

## 10.1 Data Classification

All data within the Customer Data Platform is classified according to the organization's four-tier data classification scheme: Restricted (most sensitive), Confidential, Internal, and Public. The classification drives encryption requirements, access control policies, retention periods, and cross-border transfer restrictions.

| Classification | Examples | Encryption | Access Control |
|---|---|---|---|
| Restricted | SSN, full account number, biometrics | AES-256 field-level | Named individuals only |
| Confidential | Name, DOB, address, email, phone | AES-256 at-rest | Role-based (need-to-know) |
| Internal | Account type, product holdings, branch | AES-256 at-rest | All authenticated employees |
| Public | Branch locations, product catalog | TLS in-transit only | No restrictions |

Classification is assigned at the field level within the data catalog (managed by AWS Lake Formation) and is inherited by any derived datasets that include the field. When a dataset is created that combines fields of different classifications, the resulting dataset inherits the highest classification present. This ensures that sensitive fields cannot inadvertently appear in lower-classification contexts.

## 10.2 GDPR & CCPA Compliance

The Customer Data Platform implements comprehensive controls to support compliance with the General Data Protection Regulation (GDPR), the California Consumer Privacy Act (CCPA), and equivalent state-level privacy legislation. Key capabilities include data subject access requests (DSAR), right to erasure (right to be forgotten), data portability, consent management, and purpose limitation.

Right to Erasure is the most architecturally complex privacy requirement because customer data exists across multiple storage tiers (Oracle Exadata, MSK, S3, Redshift, OpenSearch) and in multiple formats (operational records, analytical aggregations, ML model training data, backup media). The CDP implements a centralized deletion orchestrator that, upon receiving an erasure request, coordinates deletion across all storage locations within the 30-day regulatory window.

The deletion orchestrator maintains a deletion manifest that tracks the erasure status for each storage location. For mutable stores (Exadata, OpenSearch, Redshift), records are physically deleted. For immutable stores (S3 Parquet files), affected partitions are rewritten with the subject's data removed. For Kafka topics, the subject's events are overwritten using Kafka's log compaction with tombstone records. Backup media follows a crypto-shredding approach where per-customer encryption keys are destroyed, rendering the backed-up data unrecoverable.

## 10.3 Encryption & Access Control

Encryption is applied at multiple layers to provide defense-in-depth protection for customer data. Data in transit uses TLS 1.3 for all API communication, TLS 1.2 (minimum) for Kafka producer/consumer connections, and mutual TLS (mTLS) for service-to-service communication within the platform. Data at rest uses AES-256 encryption managed by AWS KMS with customer-managed keys (CMKs) that the bank controls and can revoke at any time.

Restricted-classification fields (SSN, full account numbers) receive an additional layer of field-level encryption using application-layer tokenization. The tokenization service generates format-preserving tokens that maintain the statistical properties of the original data (enabling analytics on tokenized data) while being irreversible without access to the tokenization key stored in a dedicated HSM partition. Detokenization is permitted only for authorized business processes (identity verification, regulatory reporting) and all detokenization events are logged to an immutable audit trail.

Access control follows a role-based model with principle of least privilege. IAM policies in AWS and Oracle Database Vault on-premises restrict access based on job function. The Data Engineering team can read/write to all platform components but cannot access Restricted-classified fields. The Data Science team can read curated data for model training but receives only tokenized PII. The Customer Service team can query the Customer 360 API with detokenization rights for customer-facing use cases. All access is logged and reviewed quarterly.

# 11 Monitoring & Observability

The monitoring architecture provides end-to-end visibility across the hybrid infrastructure, tracking data freshness, pipeline health, component performance, and business-level metrics. The observability stack combines Amazon CloudWatch for AWS resource metrics, Prometheus/Grafana for application-level metrics, PagerDuty for alert routing and escalation, and Oracle Enterprise Manager for on-premises database monitoring.

The primary monitoring dashboard displays the following key health indicators in real-time: CDC pipeline lag (target: < 5 seconds, current: 1.8s average), identity resolution throughput (target: > 4M records/hour, current: 4.2M), Customer 360 API P99 latency (target: < 50ms, current: 34ms), data quality score (target: > 99%, current: 99.4%), and source system feed health (14/14 active, showing last received timestamp for each).

Alert thresholds are organized into three severity tiers. Warning alerts (P3) indicate a metric approaching concerning levels and require investigation during business hours -- examples include CDC lag exceeding 30 seconds, a single source feed being stale for more than 1 hour, or identity resolution throughput dropping below 3.5M records/hour. Critical alerts (P1) indicate active or imminent service impact requiring immediate response -- examples include complete CDC pipeline failure, Customer 360 API error rate exceeding 1%, or data quality score dropping below 98%.

| Metric | Warning Threshold | Critical Threshold | Current Value |
|---|---|---|---|
| CDC Pipeline Lag | > 30 seconds | > 5 minutes | 1.8 seconds |
| Customer 360 API P99 | > 75 ms | > 200 ms | 34 ms |
| API Error Rate | > 0.5% | > 1% | 0.02% |
| Identity Resolution TPS | < 3.5M/hr | < 2M/hr | 4.2M/hr |
| Data Quality Score | < 99% | < 98% | 99.4% |
| Source Feed Staleness | > 1 hour | > 4 hours | < 5 seconds (all) |
| Kafka Consumer Lag | > 50,000 msgs | > 200,000 msgs | < 1,200 msgs |
| Redshift Query Queue | > 10 queued | > 25 queued | 2 avg |

Distributed tracing is implemented using AWS X-Ray for cloud components and Oracle Application Continuity for on-premises database operations. A unique correlation ID is assigned to each event at the point of ingestion (GoldenGate Extract) and propagated through all processing stages. This enables end-to-end latency analysis and rapid root-cause identification when pipeline performance degrades.

# 12 Backup & Disaster Recovery

The backup and disaster recovery strategy for the Customer Data Platform is designed to protect against three categories of failure: component failure (handled by HA mechanisms), site failure (handled by cross-site DR), and data corruption (handled by point-in-time recovery from backups). Each category has distinct recovery objectives and mechanisms.

| Component | Backup Method | Frequency | RTO | RPO |
|---|---|---|---|---|
| Oracle Exadata | RMAN incremental + Data Guard | Continuous + daily full | 3 min (DG) | < 15s |
| Amazon MSK | Topic replication (3x) + S3 archive | Continuous | < 30 seconds | 0 (within cluster) |
| S3 Data Lake | Cross-region replication | Continuous | < 1 minute | < 15 minutes |
| Amazon Redshift | Automated snapshots | Every 8 hours | < 10 minutes | < 8 hours (snapshot) |
| OpenSearch | Automated snapshots to S3 | Hourly | < 15 minutes | < 1 hour |
| Airflow Metadata | RDS Multi-AZ + snapshots | Continuous + daily | < 5 minutes | < 1 minute |

The most critical DR scenario is a complete loss of DC-East, which would eliminate the Oracle Exadata primary database. In this scenario, the Data Guard standby at DC-West is activated as the new primary (RTO: 3 minutes, RPO: < 15 seconds). The cloud tier continues to operate normally, switching its CDC source from the GoldenGate primary to the new primary database at DC-West. Full platform recovery from a site loss has been tested quarterly, with the most recent test (March 2026) completing in 8 minutes.

For data corruption scenarios (e.g., a bug in the identity resolution engine incorrectly merges customer records), point-in-time recovery is available at each tier. Oracle RMAN supports tablespace-level point-in-time recovery. S3 versioning enables object-level rollback. Redshift supports table-level restore from snapshots. A corruption detection and recovery runbook documents the step-by-step procedure for each scenario, with an estimated recovery time of 2-4 hours depending on the scope of affected data.

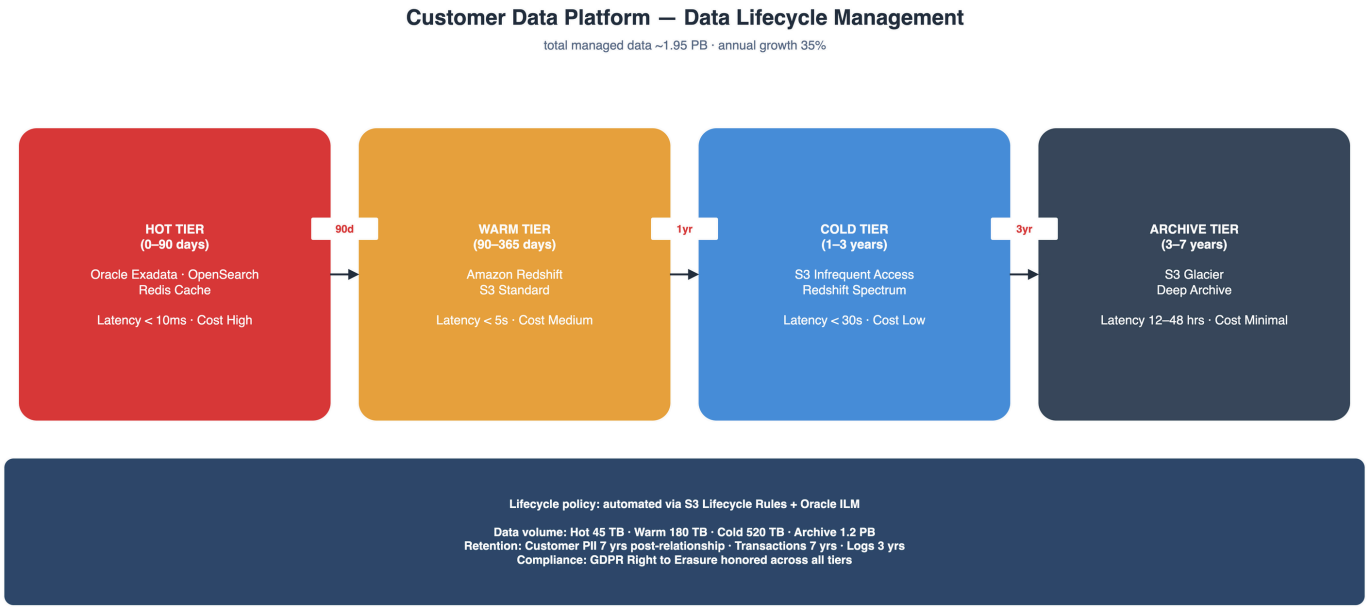



*Figure 5: Data Lifecycle Management - Storage Tiers*

Data lifecycle management ensures that aging data transitions through progressively lower-cost storage tiers according to defined policies. The hot tier (Oracle Exadata and OpenSearch) stores the most recent 90 days of data, providing sub-10ms access latency. The warm tier (Redshift and S3 Standard) stores 90-365 day data for analytical queries. The cold tier (S3 Infrequent Access) stores 1-3 year data for occasional regulatory queries. The archive tier (S3 Glacier Deep Archive) stores 3-7 year data for regulatory retention compliance.

# 13 Capacity Planning

Capacity planning for the Customer Data Platform follows a proactive model with quarterly reviews and an 18-month forward-looking forecast. The platform's hybrid nature means that on-premises capacity must be planned with longer lead times (6-9 months for Exadata expansion) while cloud capacity can be adjusted more dynamically (days to weeks for most AWS services).

| Resource | Total Capacity | Current Utilization | Projected (12 mo) |
|---|---|---|---|
| Exadata CPU (cores) | 256 | 148 (58%) | 192 (75%) |
| Exadata Storage | 200 TB | 134 TB (67%) | 175 TB (88%) |
| Exadata IOPS | 12M | 3.2M (27%) | 4.8M (40%) |
| MSK Throughput | 100 MB/s per broker | 32 MB/s (32%) | 48 MB/s (48%) |
| MSK Storage | 18 TB (9 brokers) | 8.4 TB (47%) | 12.6 TB (70%) |
| Redshift Storage | 512 TB | 180 TB (35%) | 245 TB (48%) |
| OpenSearch Storage | 6 TB | 3.8 TB (63%) | 5.1 TB (85%) |
| Direct Connect BW | 20 Gbps (2x10G) | 4.8 Gbps (24%) | 6.5 Gbps (33%) |
| S3 Data Lake | Unlimited | 745 TB | ~1 PB (projected) |

The most capacity-constrained resource is Oracle Exadata storage, projected to reach 88% utilization within 12 months at current growth rates. A storage expansion project (adding 4 additional storage cells) is approved and scheduled for Q3 2026 installation, which will increase total capacity to 320 TB and provide runway through Q4 2027.

OpenSearch storage is the second most constrained resource at 63% current utilization and 85% projected. The mitigation plan involves implementing index lifecycle management (ILM) policies that automatically move older indices to warm storage nodes (lower-cost d3.2xlarge instances) and ultimately delete indices older than 18 months that have been archived to S3. This ILM policy is scheduled for implementation in Q2 2026 and is expected to reduce hot storage utilization to below 50%.

Cloud-based resources (MSK, Redshift, S3) benefit from the ability to scale elastically without long lead times. MSK can be scaled by adding brokers (with automatic partition reassignment), Redshift can be resized via elastic resize (completing in minutes for ra3 node types), and S3 has effectively unlimited capacity. Budget approvals for cloud scaling are pre-authorized up to a 30% increase from current monthly spend, enabling the operations team to respond to capacity needs without waiting for a procurement cycle.

# 14 Known Risks & Deviations

The following risks and architectural deviations have been identified, assessed, and formally accepted through the Architecture Review Board (ARB). Each item includes the risk description, current mitigation strategy, and responsible owner.

**R-001: Single-region AWS deployment: All AWS components are deployed in us-east-1 only. A regional AWS outage would render the cloud tier (MSK, Redshift, OpenSearch, S3) unavailable. Mitigation: On-premises Oracle continues serving the Customer 360 API via direct database queries (degraded performance but functional). Cross-region replication for S3 is implemented but full multi-region for compute services is deferred due to cost. Status: Accepted (ARB-2025-089). Owner: Cloud Architecture.**

**R-002: Identity resolution ML model retraining lag: The ML ensemble model (Stage 4) is retrained monthly, creating a window where concept drift may reduce matching accuracy. Monitoring shows accuracy degradation of 0.1-0.3% by end of month before retraining restores baseline. Mitigation: Weekly accuracy monitoring with manual review escalation if accuracy drops below 99.9%. Automated retraining pipeline (reducing cycle to weekly) is in development. Status: In Progress. Owner: Data Science.**

**R-003: GoldenGate single Extract process: While GoldenGate Extract has a standby for failover, both processes run on the same physical Exadata node. A node failure requires Extract restart on another node, adding 2-3 minutes of CDC downtime. Mitigation: GoldenGate Microservices Architecture migration (planned Q4 2026) will distribute Extract across multiple nodes. Status: Planned. Owner: Database Engineering.**

**R-004: Schema registry single point of failure: The Confluent Schema Registry runs as a single ECS task with a standby. During failover, GoldenGate producers cache schemas locally and continue publishing, but new schema registrations are blocked until the registry recovers (estimated 30-60 seconds). Mitigation: Schema changes are infrequent (average 2 per week) so the window of vulnerability is small. Multi-node registry cluster planned for Q3 2026. Status: Accepted (ARB-2026-012). Owner: Data Platform Engineering.**

**R-005: Direct Connect bandwidth ceiling: At 35% annual data growth, the 20 Gbps aggregate Direct Connect capacity will be insufficient by Q2 2028. Lead time for additional circuits is 4-6 months. Mitigation: Capacity review scheduled for Q1 2027 to initiate procurement if growth projections hold. Additionally, data compression (currently not applied to CDC events) could reduce bandwidth by an estimated 60%. Status: Monitoring. Owner: Network Engineering.**

**R-006: Deletion orchestrator complexity: The GDPR deletion orchestrator must coordinate erasure across 7 distinct storage systems. Testing has identified edge cases where deletion fails silently in one system (particularly Kafka tombstone propagation during partition rebalancing). Mitigation: Daily reconciliation audit verifies deletion completeness. Any incomplete deletions are retried automatically. Manual verification for high-sensitivity erasure requests. Status: In Progress. Owner: Privacy Engineering.**

**R-007: Exadata storage approaching capacity limit: At current growth rate (35% YoY), Exadata storage will reach 88% utilization within 12 months, crossing the 85% threshold that triggers performance degradation for write operations. Mitigation: Storage expansion project approved and scheduled for Q3 2026 with 4 additional storage cells. Interim mitigation includes accelerating archival of cold data to S3. Status: In Progress. Owner: Database Engineering.**

## 14.1 Risk Mitigation Timeline

The following table summarizes the planned resolution timeline for each active risk item, including the expected completion date, investment required, and the executive sponsor responsible for ensuring delivery:

| Risk ID | Mitigation Action | Target Date | Status |
|---|---|---|---|
| R-001 | Cross-region S3 replication (completed), Compute DR eval | Q1 2027 | In Progress |
| R-002 | Automated retraining pipeline via SageMaker Pipelines | Q3 2026 | In Progress |
| R-003 | GoldenGate Microservices Architecture migration | Q4 2026 | Planned |
| R-004 | Multi-node Schema Registry cluster on ECS | Q3 2026 | Planned |
| R-005 | Direct Connect capacity expansion (2x10G additional) | Q1 2028 | Monitoring |
| R-006 | Deletion orchestrator v2 with transactional guarantees | Q2 2026 | In Progress |
| R-007 | Exadata storage expansion (4 additional cells) | Q3 2026 | Approved |

Risk acceptance decisions are reviewed annually by the Architecture Review Board to confirm that the risk profile remains acceptable given changes in the threat landscape, regulatory environment, and business requirements. The most recent annual review (February 2026) confirmed all existing acceptances and added R-007 as a new tracked risk based on updated growth projections.

## 14.2 Architectural Debt Register

In addition to the formal risk register, the platform maintains an architectural debt register that tracks technical decisions made for expediency that should be revisited. These items do not represent active risks but are areas where the current implementation deviates from the ideal architecture and may become problematic as the platform scales.

**AD-001: Monolithic Identity Resolution Engine: The identity resolution engine currently runs as a single monolithic application on dedicated Exadata nodes. As matching volume grows, this architecture limits horizontal scalability. The long-term plan is to decompose into a microservices architecture with separate services for each matching stage, enabling independent scaling. This refactoring is planned for H2 2026 but is not yet funded.**

**AD-002: Hardcoded Data Quality Rules: Approximately 30% of data quality rules are implemented as hardcoded logic in Flink operators rather than externalized to the Great Expectations rule engine. This makes rule updates require code deployments rather than configuration changes. Migration of remaining rules is proceeding incrementally with a target of full externalization by Q4 2026.**

**AD-003: Legacy JDBC Batch Connectors: Three source systems (branch teller, ATM network, legacy CRM) still use direct JDBC polling for batch ingestion rather than CDC-based capture. These connectors impose load on source systems during extraction windows and provide lower data freshness than the CDC approach. Migration to GoldenGate-based CDC for these systems is planned but depends on source system upgrade projects**

**currently in progress.**

**AD-004: Manual Scaling Procedures: While cloud components support auto-scaling, the procedures for scaling on-premises components (adding Exadata nodes, expanding GoldenGate capacity) remain manual and require significant lead time. Infrastructure-as-code tooling (Terraform for cloud, Ansible for on-premises) is being extended to cover all scaling operations, with full automation targeted for Q1 2027.**

# 15 References

- Oracle Exadata Database Machine X9M Technical Architecture (Oracle White Paper)